

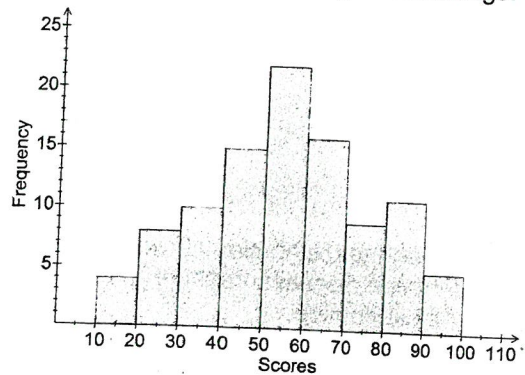
③ Interpreting Histograms.

1. Describe the distribution of the scores displayed by the histogram below under the given headings.

(a) Measures of spread:

(b) Measures of central tendency:

(c) Shape:

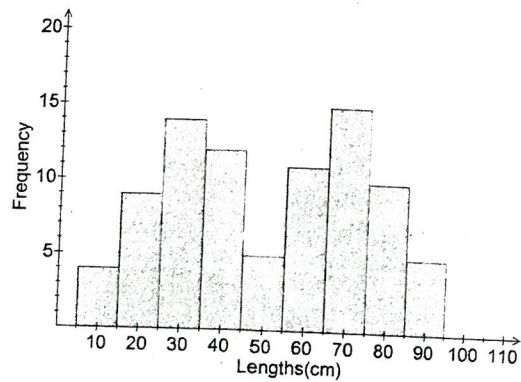


2. Describe the distribution of the lengths displayed by the histogram below under the given headings.

(a) Measures of spread:

(b) Measures of central tendency:

(c) Shape:

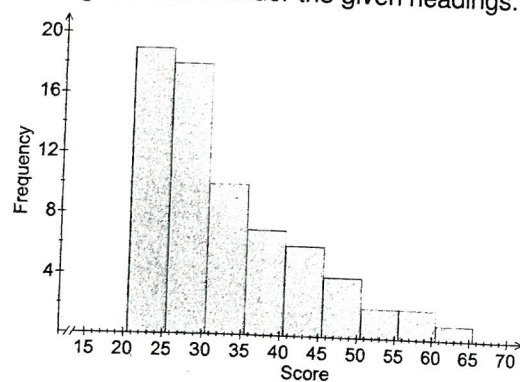


3. Describe the distribution of the scores displayed by the histogram below under the given headings.

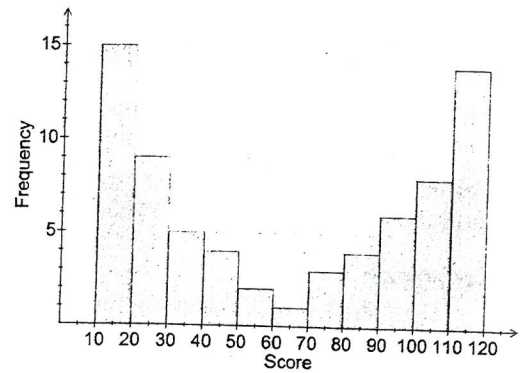
(a) Measures of spread:

(b) Measures of central tendency:

(c) Shape:



4. Describe the distribution of the scores displayed by the histogram below under the given headings.
 (a) Measures of spread:

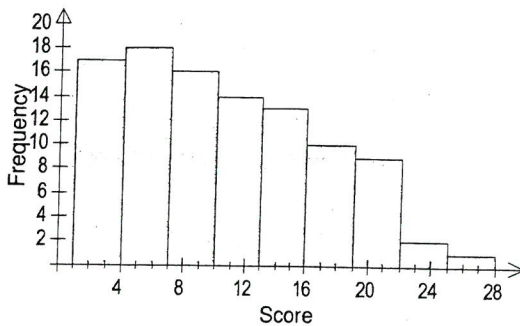


(b) Measures of central tendency:

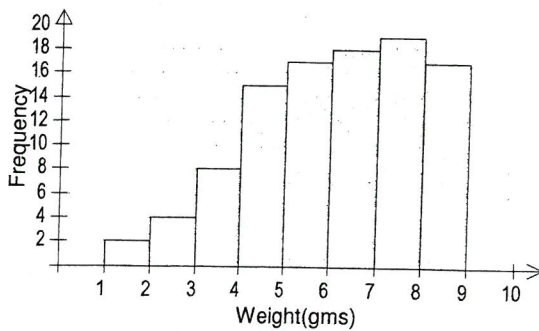
(c) Shape:

5. Describe the distribution shown in each of the following histograms labelled graphs A to F. In your description discuss: measure(s) of central tendency, measure(s) of dispersion, and the overall shape of the distribution.

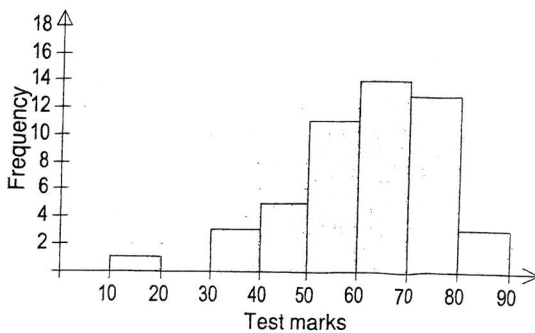
Graph A



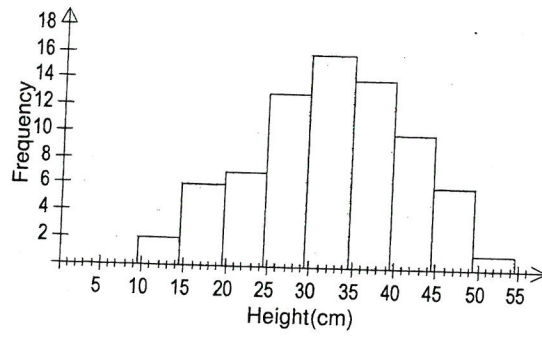
Graph B



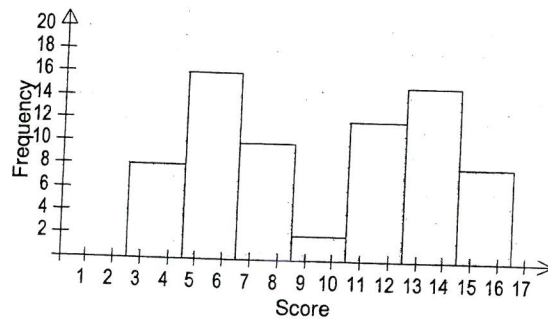
Graph C



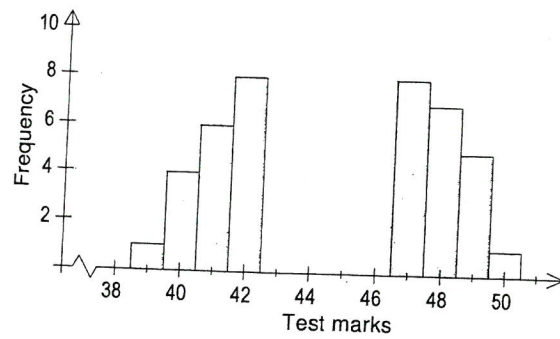
Graph D



Graph E



Graph F



Solutions ③

1. (a) The data displayed by the histogram involves 100 scores. The scores are spread between about 10 and 100 with a range of about 90. The standard deviation has been estimated to be 20.51 (2d.p.) using the class midpoints
- (b) The mean has been estimated to be 56.2 by using the class midpoints. As the total number of scores is 100 the middle score is in the range 50 to 60, that is the median class of the scores is 50 to 60 marks. The modal class is the same as the median class that is 50 to 60.
- (c) The distribution of scores is unimodal, symmetrical, peaking at the median/modal class of 50 to 60 and falling away reasonably symmetrically left and right of this central column.
2. (a) The histogram displays 85 lengths which are spread from about 5 cm to about 95 cm with a range of approx. 90 cm. The standard deviation has been estimated using the class midpoints of the columns and is 23.09 cm (2 d.p.)
- (b) The mean has been estimated to be 50.94 cm (2 d.p.) by using the class midpoints. As the total number of lengths is 85 the middle length is in the range 45 to 55, that is the median class of the lengths is 45 to 55 cm. The distribution has no unique modal class it is bi-modal with modal classes 25 to 35 cm and 65 to 75 cm.
- (c) The lengths are reasonably symmetrically spread about a midpoint of about 50 cm. As mention earlier the distribution of lengths is bimodal with peaks and clustering of the lengths about 30 cm and 70 cm. Both clusters of lengths are reasonably symmetrical.
3. (a) The histogram displays 69 scores which are spread from about 20.5 to about 65.5, a range of about 45. The standard deviation has been estimated using the class midpoints of the columns of the histogram and is 10.05 cm (2 d.p.)
- (b) The mean has been estimated to be 32.93 (2 d.p.) by using the class midpoints. As the total number of scores is 69 the middle score is in the second column that is in the interval 25.5 to 30.5, that is the median class of the scores is 25.5 to 30.5. The distribution has modal class 20.5 to 25.5.
- (c) The distribution of scores is unimodal and skewed to the right with the first interval 20.5 to 25.5 having the highest frequency with approximately 28% of the scores. Approximately 54% of all the scores lie within the first two class intervals, that is between 20.5 and 30.5. Then the number of scores in the following class intervals taper off fairly dramatically with the last three class intervals accounting for only some 7% of the scores in the distribution.
4. (a) The 71 recorded scores of the distribution are spread from about 10 to about 120 with a range of approximately 110. The standard deviation has been estimated using the class midpoints of the columns of the histogram and is 39.94 (2 d.p.)
- (b) The mean, estimated using class midpoints is 64.3 (1 d.p.) and is located approximate in the central interval 60 to 70. The

median class is the interval 60 to 70 and is located in the central interval of the distribution. The scores peak around 15 and again at 115 and the distribution can be described as bimodal.

- (c) The scores are reasonably bimodal and symmetrical about the median class or class midpoint 65. The distribution has dense regions at both ends and a very much less dense region centrally. The distribution takes on a U shape.

5. GRAPH A: The distribution has 100 scores with an estimated mean of 10.6 determined using the midpoints of the class intervals. The median lies in the interval 7 to 10 and the modal class is in the interval 4 to 7. The scores are spread from approximately 1 to 28 with a range of approximately 27. The standard deviation of the scores is 6.17 (2 d.p.) estimated using the class midpoints.

The distribution is unimodal within the 4 to 7 class interval and is skewed to the right.

Graph B: The distribution has 100 weights with an estimated mean weight of 6.06 gm (using class midpoints), the median weight lies in the 6 gm to 7 gm class interval and the modal class is from 7 gm to 8 gm interval. The weights are spread from approximately 1 gm to 9 gm with a range of approximately 8 gm and an estimated standard deviation of 1.82gm(2 dp). The distribution of the weights is skewed to the left and the distribution of the weights is unimodal. 86% of the weights are at least 4 gms and the distribution of these weights is approximately uniform.

Graph C The distribution has 50 test marks with a estimated mean mark of 61.8 (using class interval midpoints). The median lies in the 60 to 70 mark interval and the modal class is also the 60 to 70 mark class interval.

The test marks are spread from approximately 10 marks to 90 marks with a range of approximately 80 marks. The standard deviation (estimated using class midpoints) of the test marks to 2dp is 14.34.

The distribution is unimodal within the interval 60 to 70 test marks and skewed to the left with approximately 82% of the marks at least 50 and 18% of the marks below 50. The distribution has a gap with no marks in the range 20 marks to 30 marks and only 1 mark in the first class interval 10 to 20 marks.

Graph D The distribution of the 75 heights in the sample has an estimated mean of 32.2cm determined using class interval midpoints. The modal class and the median class intervals are both 29.5 to 34.5 cm. The heights are spread from approximately 9.5 cm to 54.5 cm with a range of approximately 45cm. An estimate of the standard deviation of the heights is 9.14 cm correct to 2 dp. (using class midpoints)

The distribution of the heights is unimodal and symmetrical about the 29.5 cm to 34.5 cm class interval which contains the mean, median and is the modal class. Class intervals either side of this central interval have diminishing frequencies. Approximately 70% of the heights lie in the range 24.5cm to 44.5cm.

Graph E The sample has 71 scores with an estimated mean of 9.5, (estimated using class midpoints) the modal class is 4.5 to 6.5 and the median lies in the interval 8.5 to 10.5. The scores are spread from approx 2.5 to 16.5 with a range of approximately 14. An estimate of the standard deviation of the scores is 4.04(2 d.p.) using class midpoints.

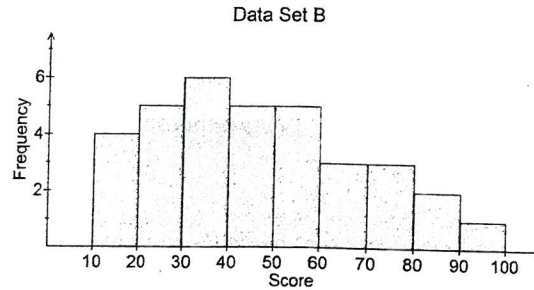
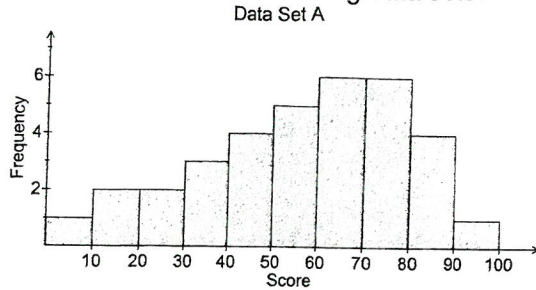
The distribution is bi-modal and symmetrical in appearance about the interval 8.5 to 10.5 with modal classes 4.5 to 6.5 and 12.4 to 14.5. Approximately 3% of the scores lie in the vicinity of the mean and median that is in the interval 8.5 to 10.5 of the remaining 97% of scores approx half are clustered around the 4.5 to 6.6 bin and the other half around the 12.5 to 14.5 bin.

Graph F: As test marks are discrete in this example we can calculate the actual statistics. The histogram displays the distribution of 40 test marks. The mean of the test marks is 44.7, median is 47 and the distribution is bimodal with modes at 42 and 47. The test marks are spread from 39 to 50 and the standard deviation of the marks is 3.54(2 d p). The distribution comprises of two clusters, i.e. bimodal, each with four bins separated by a gap also of four bin widths. The number of test marks in each cluster is approximately the same, i.e. 19 marks and 21 marks. The cluster with the lower test marks is skewed to the left and the cluster made up of the higher test marks is skewed to the right.

The test marks are distributed symmetrically about the gap. This histogram is rather unusual for test marks and it appears as if the four central bin test marks have been removed as tests marks usually follow a bell-shaped distribution.

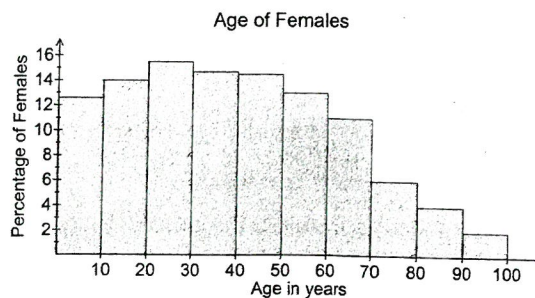
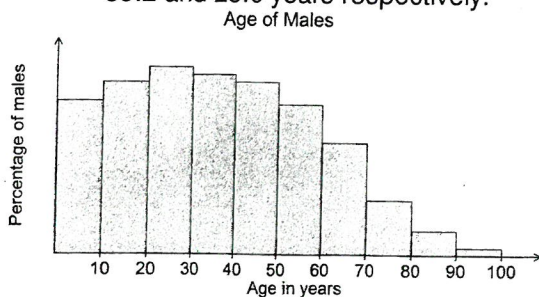
④ Comparing Histograms.

1. Consider the following data sets:



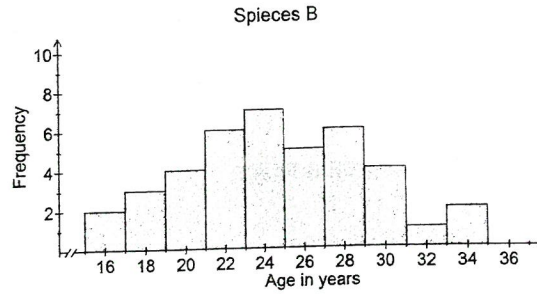
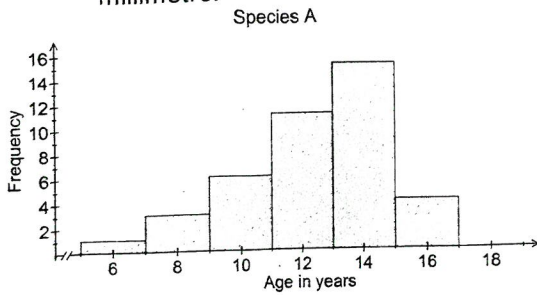
- Estimate the mean and standard deviation for both data sets.
- Describe the shape of the data for both data sets.
- Comment on the similarities and differences between data sets A and B.

2. The histograms show the distribution of the ages of West Australian males and females in the year 2000. The mean and standard deviation for the males are 37.9 and 21.1 years and for the females 39.2 and 23.0 years respectively.



Comment on the similarities and differences between data sets A and B.

3. An entomologist records the body lengths of 40 moths, of two different species A and B to the nearest millimetre.



(a) State variable for these histograms and indicate whether is continuous or discrete. Justify.

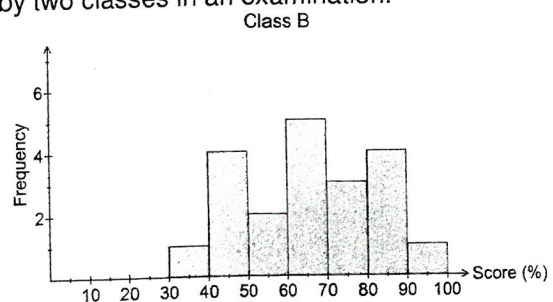
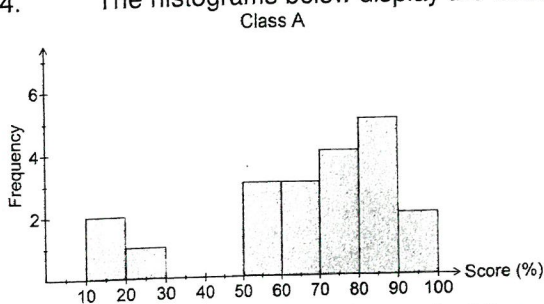
(b) The mean and standard deviation of species A moths is 12.4 mm and 2.37 mm. Estimate the mean and standard deviation of species B.

(c) Which species has the longer body length? Justify your response.

(d) Which moth species has more variation in body length? Justify your response.

(e) Compare the modality and shape of the distribution of body length for these species.

4. The histograms below display the marks achieved by two classes in an examination.



Without using a calculator select the correct answer for parts (a) to (d):

(a) Which class has the larger range?
Class A Class B Neither

(b) Which class has the larger standard deviation?
Class A Class B Neither

(c) Which class has the higher median?
Class A Class B Neither

(d) Which class has the higher mean?
Class A Class B Neither

(e) Comment on any similarities or differences between the exam results for these classes.

Solutions ④

1. (a) Mean A = 56.47 (2 d.p.) Std dev'n A = 22.51 (2 d.p.)
 Mean B = 46.47 (2 d.p.) Std dev'n A = 21.85 (2 d.p.)
 (b) Data set A: The distribution is unimodal over the interval 60 to 80 with just over 35% of scores in this group. The scores

are clustered about 70 and their distribution is negatively skewed

Data set B: The distribution is unimodal over the interval 30 to 40. The scores are clustered about the modal score interval with approx. 74% of the scores in this cluster between 10 and 60.. The distribution of the scores for this data set is positively skewed.

(c) Both data sets have 34 data points.

The mean for data set A is significantly greater than the mean for data set B, 56.47 compared to 46.47 indicating that the scores for set A are much higher than those for set B.

The scores for data set A are slightly more spread than those of data set B as indicated by standard deviations of 22.51 and 21.85 respectively and the observation that the scores for A range from 0 to 100 and those for B range from 10 to 100. Both distributions are skewed, distribution A is skewed to the left and distribution B is skewed to the right.

2. The mean age for females is slightly higher than that for the males, 39.2 year compared to 37.9 years, indicating that females tend to live longer than males. This is also evident when visually comparing the graphs where the percentages in the latter years for females are distinctly higher. The ages of the females are more spread than those of males as evident by comparing their standard deviations, that is 23.0 compared to 21.1 years.

Both male and female age distributions are positively skewed as the number and hence percentage of older people decreases with age.

3. (a) Variable is the body length of moths and it is a continuous variable as the length can take on any value in a given interval. (b) Mean = 24.6 mm, std dev'n = 4.57 mm.

(c) Species B has the longest body length because the mean body length of species B is 24.6mm compared to only 12.4 mm for species A.

(d) Species B has more variation in body length with standard deviation of 4.57 mm compared to only 2.37 mm for species A.

(e) Both distributions are unimodal, species A has a modal length of 13 to 15 mm and species B 23 to 25 mm. The distribution of body lengths for species A is negatively skewed and the distribution of body lengths for species B is approx. symmetrical.

4. (a) Class A (b) Class A (c) Class A (d) Class A

(e) Class is unimodal, the modal interval of class

A is the interval 80 to 90 marks. Class B may be described as being multi-modal as it has 3 distinct peaks within the following intervals, 40 to 50, 60 to 70 and 80 to 90.

Class A's distribution of marks tends to show some small degree of skewness to the left whereas the distribution of marks for class B is approximately symmetrical about the modal interval 60 to 70 marks.

Class A's distribution of marks have a gap with no marks recorded between 30 and 50. Hence the display for class A shows a small cluster of 3 student marks in the interval 10 to 30 and the remainder in a large cluster from 50 and 100